

4.16 Revised ERD

Submission by Steve Gonzales for IMT543

In this evolution of the ERD, I introduced user management system, making the schema multi-tenant capable rather than single-user focused. The addition of budget, expense, income, and savings_goal tables transforms the system from simple transaction logging to comprehensive financial planning.

This evolution employs enhanced data modeling patterns like:

- Proper normalization: Better separation of concerns
- Constraint-driven design: Business rules enforced at database level
- Scalable architecture: Supports complex financial scenarios

The revised ERD for my Project includes these added tables:

- Savings Goal. This allows the user to enter financial goals that can then track progress. E.g. if you were saving to put a down payment on a house.
- Budget. This allows the user to have set and follow a spending budget. It will include total monthly amounts and actuals can then be tracked against the budget.
- Expense. This is similar to the transaction table, but will only include expenses for the purpose of tracking the budget.
- Income. Similar to Expenses, this will track income coming in.

